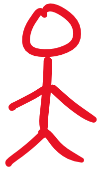
RATIO



Milk = 1 L

Aloo = 2 kg

$$\frac{M}{A} = \frac{1}{2}$$

Ratio can't be found

Ratio is unitless

$\downarrow$

Ratio is taken when units are same

$\downarrow$

A = 5 marks

V = 8 marks

$$\Rightarrow \frac{A}{V} = \frac{5}{8}$$

① $\frac{A}{B} = \frac{1\cancel{00}\,\cancel{M}}{2\cancel{00}\,\cancel{M}} = \frac{1}{2}$

$\frac{A}{B} = \frac{1}{2}$

$A = 1$, $B = 2$ } May / may not be True

$\frac{A}{B} = \frac{k}{2k}$

$A = k$, $B = 2k$ } Always True //

9]

$$\frac{P}{C} = \frac{1}{2}$$

Sum of marks of P & C is 200 M.

Marks of P & C is ?

Sol^n : ① $\frac{P}{C} = \frac{k}{2k}$

M1 → $P = k$ ⟶ 200/3 marks

$C = 2k$ ⟶ 400/3 marks

② Sum of marks of P & C is 200 M.

$$k + 2k = 200$$

$$3k = 200$$

$$\boxed{k = \frac{200}{3}}$$

Ratio as part of Total value

① $\frac{A}{B} = \frac{p}{q}$

② → $A = pk$

$B = qk$

③ → $A + B = pk + qk = (p+q)k$

$$\boxed{\therefore\ k = \frac{A+B}{p+q}}$$

$$A = pk = \frac{p}{p+q} \cdot (A+B)$$

$$B = qk = \frac{q}{p+q} \cdot (A+B)$$

$$\therefore \boxed{k = \frac{A+B}{p+q}}$$

$A+B$ → Total value

9]

$$\frac{P}{C} = \frac{1}{2}$$

Sum of marks of P & C is 200 M.

marks of P & C is ?

M2

$$P = \frac{1}{1+2} \cdot (200\,M) = \frac{200}{3}\,M$$

$$C = \frac{2}{1+2} \cdot (200\,M) = \frac{400}{3}\,M$$

Two numbers are in the ratio who is 2:3 ,if 9 is added to each number , they will be in the ratio 3:4 what is the product of two numbers?

Soln :

1. $\frac{A}{B} = \frac{2}{3}$

$\boxed{A = 2k}$, $\boxed{B = 3k}$

2. $\frac{2k+9}{3k+9} = \frac{3}{4}$

$\rightarrow k = 9$ //

3. $A \cdot B$

$= (2k)(3k)$

$= 6k^2$

$= 6(9)^2$

$= 486$ //

If x:y =7:5, then what is the value of (5x-2y):(3x+2y)?

Soln :

$$\frac{5x-2y}{3x+2y} = \frac{5(7k)-2(5k)}{3(7k)-2(5k)} = \frac{25}{31}$$

M1

$$\frac{x}{y} = \frac{7}{5} \rightarrow \begin{array}{l} x = 7k \\ y = 5k \end{array}$$

If $\boxed{x:y = 7:5,}$ then what is the value of (5x-2y):(3x+2y)?

M2

$$\frac{(5x-2y)/y}{(3x+2y)/y}$$

$$= \frac{5\left(\frac{x}{y}\right)-2}{3\left(\frac{x}{y}\right)+2} = \frac{5\cdot\frac{7}{5}-2}{3\cdot\frac{7}{5}+2}$$

$$= 25/31$$

The ratio of the income of Amit and Sumit is 5:8 and there expenditure is in the ratio of 11:10 if Amit and Sumit saves Rs 7900 and Rs 21000 respectively then their respective incomes are?

Sol^n :

Saving = Income − Expense

① $A(In) = 5k_1$

$S(In) = 8k_1$

② $A(Ex) = 11k_2$

$S(Ex) = 10k_2$

③ $\boxed{\text{Amit}} \rightarrow 7900 = 5k_1 - 11k_2$

Sumit $\rightarrow 21000 = 8k_1 - 10k_2$

$$10(7900) = (5k_1 - 11k_2)\,10$$

$$\ominus \quad 11(21000) = (8k_1 - 10k_2)\,11$$

$$k_1 = 4000$$

$$A\,(In) = 5k_1 = 5[4000] = 20,000$$

$$S\,(In) = 8k_1 = 8[4000] = 32,000$$

Ratio of more than 2 members

Q] $\frac{A}{B} = \frac{5}{2}$ $\frac{B}{C} = \frac{4}{5}$

$A : B : C = ?$

Soln :

$\frac{A}{B} = \frac{5 \times 2}{2 \times 2}$ $\frac{B}{C} = \frac{4}{5}$

$\frac{A}{B} = \frac{10}{4}$

$A : B : C$

$= 10 : 4 : 5$ //

∴ 9]

$$\frac{A}{B} = \frac{3}{4}$$

$$\frac{B}{C} = \frac{5}{7}$$

$$\frac{C}{D} = \frac{8}{9}$$

(i) $\frac{A}{C} = ?$ $= \frac{A}{B} \cdot \frac{B}{C} = \left(\frac{3}{4}\right)\left(\frac{5}{7}\right) = \frac{15}{28}$ //

(ii) $A : B : C : D$

$\therefore \quad \frac{A}{B} = \frac{3}{4} \qquad \frac{B}{C} = \frac{5 \times (4/5)}{7 \times (4/5)} = \frac{4}{28/5}$

$A : B : C = 3 : 4 : \frac{28}{5}$

$\frac{B}{C} = \frac{4}{\frac{28}{5}} \qquad \frac{C}{D} = \frac{8 \times \frac{1}{8} \cdot \frac{28}{5}}{9 \times \frac{1}{\cancel{8}_2} \cdot \frac{\cancel{28}^{7}}{5}} = \frac{28/5}{63/10}$

$A : B : C : D$

$= 3 : 4 : \frac{28}{5} : \frac{63}{10}$

$= 30 : 40 : 56 : 63$ //

Proportion

① $\frac{a}{b} = \frac{c}{d}$ [Proportion]

$$\boxed{ad = bc}$$

② $\frac{a}{b} = \frac{b}{c} = \frac{c}{d} = \frac{d}{e}$ [Continued proportion]

Property : $\frac{a}{b} = \frac{c}{d}$

① Componendo $\rightarrow$ $\frac{a+b}{b} = \frac{c+d}{d}$

② Dividendo $\rightarrow$ $\frac{a-b}{b} = \frac{c-d}{d}$

③ Componendo – Dividendo $\rightarrow$ $\frac{a+b}{a-b} = \frac{c+d}{c-d}$

4. <u>Invertendo</u> : $\frac{b}{a} = \frac{d}{c}$

5. <u>Alternendo</u> : $\frac{a}{c} = \frac{b}{d}$

9]

$$\frac{a-b}{a+b} = \frac{1}{5}$$

Find $\dfrac{a^2-b^2}{a^2+b^2}$

Soln: 1 $\dfrac{a-b}{a+b} = \dfrac{1}{5}$

M1

$$\frac{(a-\cancel{b})+(a+\cancel{b})}{(\cancel{a}-b)-(\cancel{a}+b)} = \frac{1+5}{1-5}$$

$$\frac{\cancel{2}a}{-\cancel{2}b} = \frac{\cancel{6}\,3}{-\cancel{4}\,2}$$

$$\boxed{\frac{a}{b} = \frac{3}{2}}$$

$$\frac{a^2-b^2}{a^2+b^2} = \frac{\frac{a^2}{b^2}-1}{\frac{a^2}{b^2}+1}$$

$$= \frac{\frac{9}{4}-1}{\frac{9}{4}+1}$$

$$= \frac{5}{13} \,//$$

9]

$$\frac{a-b}{a+b} = \frac{1}{5}$$

find $\dfrac{a^2-b^2}{a^2+b^2} = \dfrac{3^2-2^2}{3^2+2^2} = \dfrac{5}{13}$ //

M2

[Substitution]

$a=3$

$b=2$

$$\left\{ \boxed{\frac{a-b}{a+b} = \frac{1}{5}} \right.$$

Variation

\# 1 $\boxed{\uparrow \text{Study}, \uparrow \text{mrks}}$

2 $\uparrow$ gf, $\downarrow$ Career

1) Direct proportion $\Rightarrow A \propto B$

$\Rightarrow A = kB$

2) Inverse proportion $\Rightarrow A \propto 1/B$

$\Rightarrow A = k/B$

Two alloys A and B contain gold and copper in the ratio of 2:3 and 3 :7 by mass, respectively. Equal masses of alloys A and B are melted to make an alloy C. The ratio of gold to Copper in alloy C is

(A)5:10 (B)7:13 (C)6:11 (D)9:13 [EC 2018]

Sol^n :

① Assume : A = B = 1 kg

② Alloy A : $\frac{g_A}{c_A} = \frac{2}{3}$

$\rightarrow g_A = \frac{2}{2+3} \cdot 1\,kg = \frac{2}{5}\,kg$

Alloy A : $\dfrac{q_A}{c_A} = \dfrac{2}{3}$

$\rightarrow q_A = \dfrac{2}{2+3} \cdot 1\,kg = \dfrac{2}{5}\,kg$

$\rightarrow c_A = \dfrac{3}{2+3} \cdot 1\,kg = \dfrac{3}{5}\,kg$

Alloy B : $\frac{g}{c} = \frac{3}{7}$

$\rightarrow g_B = \frac{3}{3+7} \cdot 1\,kg = \frac{3}{10}\,kg$

$\rightarrow c_B = \frac{7}{7+3} \cdot 1\,kg = \frac{7}{10}\,kg$

Alloy C : [mix A & B]

$$g_C = g_A + g_B = 2/5 + 3/10 = 7/10 \text{ kg}$$

$$C_C = C_A + C_B = 3/5 + 7/10 = 13/10 \text{ kg}$$

$$\frac{g_C}{C_C} = \frac{7/10}{13/10} = \frac{7}{13}$$

The price of a wire made of a super alloy material is proportional to the square of its length. The price of 10 m length of the wire is rupees 1600. What would be the total price in Rupees of two wires of length 4 m and 6 m?
(A)768 (B)832 (~~C~~)1440 (~~D~~)1600 [CE 2018]

$l_1 = 4m$

$l_2 = 6m$

M1

① $P \propto l^2 \Rightarrow \boxed{P = K \cdot l^2}$

② $1600 = K \cdot (10)^2 \Rightarrow K = 16$

③ $P_1 = (16)\, l_1^2 = 16 \cdot (4)^2 = 256$

$P_2 = (16)\, l_2^2 = (16)(6)^2 = 576$

Total Price
$= 256 + 576$
$= 832$

The price of a wire made of a super alloy material is proportional to the square of its length. The price of 10 m length of the wire is rupees 1600. What would be the total price in Rupees of two wires of length 4 m and 6 m?
(A)768 (B)832 (C)1440 (D)1600 **[CE 2018]**

$P \propto l^2$

$P = K l^2$

$$\boxed{\frac{P_{I}}{P_{II}} = \frac{\cancel{K} l_{I}^2}{\cancel{K} l_{II}^2}}$$

M2

$$\frac{P_1}{P_T} = \frac{l_1^2}{l_T^2}$$

$$\frac{16\cancel{00}}{P_T} = \frac{(10)^2}{l_T^2} = \frac{1\cancel{0}^2}{(4^2+6^2)} \Rightarrow P_T = 832$$

Three friends, R,S and T shared toffee from a bowl. R took 1/3rd of the toffees but returned four to the bowl. S took 1/4th of what was left but returned three toffees to the bowl. T took half of the remainder but returned two back into the bowl. If the bowl had 17 toffees left, how many toffees were originally there in the bowl?

(A)38 (B)31 (C)48 (D)41 [EC 2011]

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× × ✓ ×

If Log(P)=1/2Log(Q)=1/3Log(R), then which of the following options is TRUE?

(A) $P^2 = Q^3R^2$

(B) $Q^2 = PR$

(C) $Q^2 = R^3P$

(D) $R = P^2Q^2$ **[CE/CS/ME 2011]**

$\log_{10}$

$\cancel{10^{\log_{10} Q}}$ $Q^{1/2}$

$\log P = \log Q^{1/2} = \log R^{1/3} = K$

$\log P = K \rightarrow P = 10^K$

$\log Q^{1/2} = K \rightarrow Q^{1/2} = 10^K \rightarrow Q = 10^{2K}$

$\log R^{1/3} = K \rightarrow R^{1/3} = 10^K \rightarrow R = 10^{3K}$

Raju has 14 currency notes in his pocket consisting of only Rs 20 notes and Rs 10 notes. The total money value of the notes is Rs 230. The number of Rs 10 notes that Raju has is

(A)5 (B)6 (C)9 (D)10 [EC/EC 2012]

①

Value	No. of notes
₹ 20	x
₹ 10	y

② $20(x+y) = (14)\times 20$

$20x + 10y = 230$

$y = 5$

Consider the following gold articles P, Q, R, S and T with different weights:

- P weighs twice as much as Q
- Q weighs four and a half times as much as R
- R weighs half as much as S
- S weighs half as much as T
- T weighs less than P but more than R

Article T will be lighter in weight than

(a) P and S
(b) P and R
(c) P and Q
(d) Q and R

$P = 2Q = 2.25x$

$Q = 4.5R = 1.125x$

$R = \underline{0.5S} = 0.25x$

$S = 0.5T$

$= 0.5(x)$

$\boxed{T = x}$

	P	Q	R	S	T
$4 \times$	$2.25x$	$1.125x$	$0.25x$	$0.5x$	x
$=$	$9x$	$4.5x$	x	$2x$	$4x$

Consider the following gold articles P, Q, R, S and T with different weights:

- P weighs twice as much as Q
- Q weighs four and a half times as much as R
- R weighs half as much as S
- S weighs half as much as T
- 1 T weighs less than P but more than R

Article T will be lighter in weight than

(a) P and S x

(b) P and R

(c) P and Q

(d) Q and R

ML

$S = \frac{1}{2}T \rightarrow T = 2S \quad T > S$